Yu-Hsuan Tseng **AI Engineer, AI Technical Manager**

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SUMMARY

Cross-disciplinary **AI Imaging** professional with 8 years of experience spanning hands-on development and technical leadership. Proven track record in orchestrating the full **Product Life Cycle (PLC)**—from ISP calibration to multimodal AI productization. Specialized in accelerating delivery (e.g., **83% reduction in PLC time** for Meitu) and optimizing high-throughput pipelines via the **NVIDIA Metropolis stack**. Patent owner; POSCO Annual R&D 1st Prize; TREC 2nd Prize. Trilingual: Chinese, English, and Korean.

WHY I FIT

End-to-End PLC Leadership: I maintain total oversight of the imaging chain, from sensor-level physics to edge-AI deployment. At Meitu, I orchestrated the production release of **key functional updates** for a customer-critical multimodal feature, compressing a 6-month estimate into a 1-month release—an **83% reduction in PLC time**.

Hands-on NVIDIA Metropolis Competence: I possess the technical depth to build and optimize industrial AOI pipelines using **TensorRT and DeepStream 9.0**. Beyond execution, I diagnosed a 95% mAP noise-induced collapse and established mitigation paths through physics-based noise-model training.

ISP-Centric Robustness & Troubleshooting: I identify the sensor-level root causes behind edge-case failures. By leveraging deep ISP domain knowledge, I can preemptively mitigate real-world artifacts—such as **jello effects (rolling shutter), sensor noise interference, and color shifts**—essential for high-precision Metropolis manufacturing workflows.

Strategic Hardware-Aware Architecture: I integrate hardware constraints directly into the planning phase. I architected a pipeline integrating **RAW-domain data into physics-grounded digital twins**, achieving **universal AWB consistency** across diverse device models (multiple iPad generations) and environments, ensuring high-fidelity rendering that is hardware-independent.

Structured Technical Governance: I pioneer the use of **Architecture Decision Records (ADRs)** to document and justify critical engineering trade-offs, such as TRT 10.14 tactic coverage gaps, ensuring that project milestones are reproducible, auditable, and strategically sound.

TECHNICAL SKILLS

Technical Program Management: PLC Lifecycle Management, Technical Governance (ADR), Stakeholder Alignment (Engineering/Hardware/Commercial/Global Brands).

NVIDIA Metropolis Stack: TensorRT 10.x (INT8/FP16 Calibration, Tactic-gap Diagnosis), DeepStream 9.0 (Multi-stream Architecture), ONNX (Dynamic Batch), MVTec AD Benchmarks.

Imaging Science & ISP: ISP Tuning (Noise, Texture, Alignment), Automated IQ Regression, Color Science (CIELAB, Universal AWB), Industrial AOI Defect Detection, VLM.

Productization & Cloud: Serverless Architecture (Alicloud Deployment), Edge-AI Deployment Workflows.

Programming & Languages: Python, PyTorch, C++, C#, MATLAB. CN (native), EN (fluent), KR (advanced).

EXPERIENCE

AI Engineer, Freelance 04/2026 - Present, Taipei, Taiwan

- NVIDIA Metropolis Manufacturing Sprint: Developed an end-to-end industrial AOI defect-detection pipeline integrating TensorRT 10.x (INT8/FP16) and DeepStream 9.0. Orchestrated optimization workflows including INT8 Entropy Calibration, L4/T4 parity verification, and tactic-gap diagnosis for YOLOv8-seg. Achieved 465 img/s aggregate throughput, providing 3.8x headroom over standard AOI SLA.

- ISP-Aware Robustness Diagnosis: Leveraged ISP fundamentals to identify critical deployment risks. Diagnosed a 95% mAP collapse triggered by production-line sensor noise (SNR 30-40 dB) and established a physical noise-model training transform to bridge the gap between quantization design and field robustness.
- Strategic Digital Twin Architecture: Architected a physics-grounded makeup rendering digital twin by integrating RAW-domain processing with a Kubelka-Munk optical model. This proactive approach ensured universal AWB performance and color accuracy across diverse device models and environmental lighting scenarios.
- Scalable Cloud Orchestration: Shipped an end-to-end serverless recommendation chain on Alicloud Function Flow (11 functions / 3 flows); maintained 100% cloud-standalone parity via automated combinatorial testing and regression fixtures.
- Applied Research in Learning-from-Video (LfV): Developed a differentiable appearance engine driven by a hybrid (physics + learned) forward renderer. Implemented an Inverse Dynamics Model (IDM) to recover motion and region data from tutorial-video datasets at scale via pseudo-labeling.
- Program Governance & Stakeholder Alignment: Managed a multi-phase delivery roadmap featuring 8 Architecture Decision Records (ADR) to document engineering trade-offs. Coordinated directly with brand PMs and business teams to align technical specifications with commercial objectives.

AI Technical Manager, Evelab Insight (Meitu Group) 03/2025 - 04/2026, Taipei, Taiwan

- Accelerated PLC Delivery: Managed the production release of key functional updates for a customer-critical multimodal AI feature. Compressed delivery from 6 months to 1 month (83% reduction) by defining agile milestones, establishing risk registers, and personally conducting algorithm prototyping.
- Operational Governance & Scalability: Instrumented automated report-quality validation via an internal dashboard, utilizing confidence scores to reduce manual review workload by 93% (from 24 mins/report to a <10% sampling rate).
- Customer-Centric Requirement Engineering: Led on-site technical interviews at international brand HQs to convert user pain points into feature specifications; post-launch data reported a 20% sales lift across adopting customers.
- Strategic Business Expansion: Designed a counter-facing "skin-improvement narrative" for Tier-1 brands, which catalyzed successful inbound inquiries from competing international labels.

Independent Technical Research & Professional Upskilling 04/2024 - 03/2025, Taipei, Taiwan

- Technical Deep-Dive: Refreshed ISP and multi-modal ML/DL frameworks; revised a first-author paper on satellite road extraction for resubmission.
- Audio ML Project: Developed an audio classifier using spectral feature analysis and KMeans clustering for label-outlier detection.

Image Signal Processing Engineer, Qualcomm 04/2022 - 04/2024, Taipei, Taiwan

- ISP Calibration & Validation: Owned ISP module calibration, tuning, and image-quality (IQ) validation across mobile and PC camera platforms shipped to global production releases; contributed to 70% frame-alignment improvement: a 2-week root-cause analysis of long-standing low-light alignment failures, characterizing reproduction conditions (illumination level, local-motion type, motion-area ratio) and supporting feature owner with tuning and validation.
- Tuned dynamic range, texture, and noise modules; refined Autofocus and EIS calibration workflows; published weekly IQ status reports across 4+ OEM customer programs.

CV Research Fellow, AI Research Center 01/2021 - 04/2022, Taipei, Taiwan

- Domain Generalization for Industrial AI: Proposed an attention-based loss function and Siamese network architecture to enhance model performance under single-domain training constraints, addressing the challenge of scarce in-domain labels in industrial environments.
- Geospatial Data Engineering: Established comprehensive post-disaster remote sensing datasets utilizing ArcGIS and the GDAL library; chaired weekly project synchronizations to align research goals with engineering execution.
- Project Coordination: Chaired weekly project synchronizations to align research goals with engineering execution. Developed road-extraction algorithms for satellite imagery, resulting in a first-author paper.

NLP Research Assistant, CFDA Lab 01/2020 - 12/2020, Taipei, Taiwan

- Award-Winning BERT-based NLP Development: Co-designed a semantic query-expansion model to resolve ellipsis in multi-turn conversational AI. Benchmarked and optimized the architecture against a massive 30M-passage corpus, securing the TREC 2020 CAsT 2nd Prize.
- Hybrid Recommendation Systems: Engineered a cross-domain VOD recommendation system integrating long/short-term user preferences; utilized Design of Experiments (DOE) for behavioral analysis and mitigated "cold-start" challenges via NLP feature extraction.

Information Modeling Engineer & Platform Lead, IDM Co. 01/2018 - 01/2020, Anyang, South Korea

- Patented Technical Innovation: Engineered a patented, zero-install PDF-based architecture for digital drawings, enabling sophisticated construction management and 3D visualization within standard PDF readers without specialized software overhead.
- Cross-Discipline Engineering Leadership: Led a 3-engineer team in a professional Korean-language environment to build a real-time collaborative 3D/structural analysis platform. Chaired bi-weekly technical reviews with structural engineers lacking software backgrounds to align physical engineering requirements.
- End-to-End Product Delivery: Owned the full release plan from technical-feasibility analysis to on-site production deployment. Orchestrated three rounds of on-site user iterations with construction managers to refine zero-install workflows.
- High-Impact Results: Awarded the POSCO 2019 Best Annual R&D Project (1st Prize) for a solution that achieved an estimated 40% reduction in labor days.

HONORS, PUBLICATIONS & EDUCATION

- TREC 2020 CAsT — 2nd Prize, Conversational Assistance Track
- POSCO 2019 — 1st Prize, Best Annual R&D Project
- Patent (2019): PDF-based Digital Drawing for Construction Management
- M.S. & B.S. in Civil Engineering, National Taiwan University.
- Exchange Program, Construction & ML Lab, Seoul National University